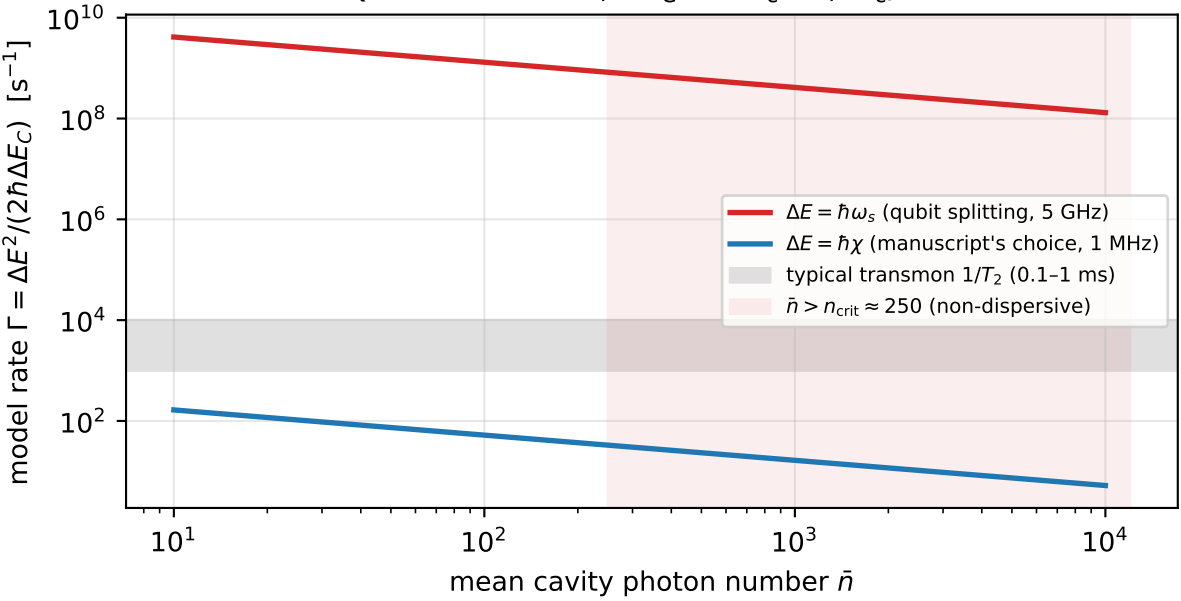
Circuit-QED model rate, $\Delta E_C = \hbar\omega_c\sqrt{\bar{n}}$, $\omega_c/2\pi = 6$ GHz
model rate $\Gamma = \Delta E^2/(2\hbar\Delta E_C)$ [s^{-1}]
10^{10}
10^{8}
10^{6}
10^{4}
10^{2}
10^{1}
10^{2}
10^{3}
10^{4}
mean cavity photon number $\bar{n}$
$\Delta E = \hbar\omega_s$ (qubit splitting, 5 GHz)
$\Delta E = \hbar\chi$ (manuscript's choice, 1 MHz)
typical transmon $1/T_2$ (0.1–1 ms)
$\bar{n} > n_{\mathrm{crit}} \approx 250$ (non-dispersive)